Additional Architecture and Governance Requirements

1. Hidden State and Information-Display Policy

The simulation engine may internally track variables such as:

- Stress
- Fatigue
- Trust
- Reputation
- Confidence
- Cognitive load
- Task backlog
- Accuracy
- Risk exposure
- Relationship quality
- Skill evidence

These variables should not automatically be displayed to the user as numerical meters.

Avoid interfaces such as:

```
Manager Trust: 72%
Communication: +5
Stress: 43%
```

Why these variables should remain hidden

Visible assessment variables encourage users to optimize the simulation rather than respond naturally.

For example, if users learn that asking a coworker for help increases a collaboration score, they may repeatedly choose that action to improve the score rather than because it reflects how they would genuinely behave.

This weakens the quality of behavioral evidence.

Hidden variables also reduce:

- Social-desirability behavior
- Score chasing
- Reverse-engineering of the assessment model
- False perceptions of psychometric precision

Three information categories

The system should classify information into three visibility levels.

A. Operationally visible information

Information the simulated professional would realistically know:

- Time
- Patient vital signs
- Number of open tickets
- Unread messages
- Budget remaining
- Task deadlines
- Equipment status
- Work queue size

This should generally be visible.

B. Experientially communicated information

Internal state that should be communicated through the world rather than displayed numerically:

- A manager becoming less patient
- A colleague becoming less willing to assist
- The user character rereading information due to fatigue
- Notifications accumulating
- Environmental noise increasing
- More frequent mistakes under workload
- Reduced clarity after prolonged concentration

This gives users meaningful feedback without revealing the assessment model.

C. Assessment-only information

Information used internally by the evidence engine:

- Skill-signal accumulation
- Confidence calculations
- Relationship scores
- Inference weights
- Cross-session pattern strength

This should remain hidden during the simulation and only appear in debriefs through evidence-based, non-numerical language.

Debrief exception

The debrief may explain the observable consequences of hidden state.

For example:

> Your manager became less willing to provide detailed guidance after several updates were delayed.

It should not normally say:

> Manager trust decreased by 18 points.

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2. Reflection Reliability and Behavioral Mismatch Detection

User reflection is valuable, but it should not be treated as unquestionable ground truth.

Users may:

- Answer in ways they think the platform expects
- Attempt to steer recommendations toward a preferred career
- Avoid admitting that an activity was difficult
- Report enjoyment because they performed well
- Report dislike because they performed poorly
- Misremember how they behaved
- Confuse satisfaction from success with enjoyment of the work itself

The evidence engine should compare reflection with behavioral and outcome evidence.

Example mismatch

The user reports:

> “The client interaction was energizing.”

However, the behavioral record shows that the user:

- Delayed opening the client message
- Avoided responding
- Delegated the conversation whenever possible
- Spent minimal time engaging with the client
- Reported relief when the interaction ended

The system should not automatically determine that either the reflection or the behavior is wrong.

Instead, it should identify the inconsistency.

Recommended debrief language

> You described the interaction as energizing, although you delayed engaging with it several times. This may reflect uncertainty, interface difficulty, initial nervousness, or a difference between anticipating an interaction and experiencing it. More situations would be needed before drawing a conclusion.

Mismatch categories

The evidence engine should be able to identify:

- Reported enjoyment versus behavioral avoidance
- Reported confidence versus repeated help-seeking
- Reported dislike versus persistent voluntary engagement
- Reported low stress versus rushed or error-prone behavior
- Reported interest versus rapid disengagement
- Reported boredom versus extensive optional exploration

Important limitation

Behavior should not automatically override self-report.

Avoidance may result from:

- Interface confusion
- Disability-related barriers
- Language difficulty
- Lack of prerequisite knowledge
- Fear of making a simulated mistake
- Misunderstanding the task

The system should treat mismatches as questions requiring further evidence, not as proof that the user is being dishonest or lacks self-awareness.

Reflection design

To reduce strategic reporting, reflection questions should be:

- Specific to a recent event
- Separated from career recommendations
- Framed without an obvious desirable answer
- Asked in different forms across sessions
- Optional when they involve sensitive emotional information

Instead of asking:

Did you enjoy communication?

Ask:

Which part of the client conversation required the most effort?

and:

Would you want interactions like this several times each day?

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3. Safety-Stack Cost and Latency Budget

The safety and consistency pipeline must be designed as an optimized system rather than a chain of several large-model calls for every dialogue line.

A naïve implementation could require:

1. Input moderation
2. Intent classification
3. NPC-context filtering
4. Dialogue generation
5. Knowledge-leakage detection
6. Contradiction checking
7. Professional-rule validation
8. Output moderation

Running separate large language models for all eight steps would create unacceptable cost and latency.

Recommended layered approach

Layer A: Deterministic checks

Use rules, schemas, and access-control logic for:

- NPC fact visibility
- Available actions
- Restricted state transitions
- Professional-rule enforcement
- Known prohibited outcomes
- Structured-output validation

These checks should not require an LLM.

Layer B: Lightweight classifiers

Use small, fast models for:

- Input moderation
- Off-topic detection
- Harassment detection
- Prompt-injection detection
- Output safety classification
- Simple fact-reference validation

These checks may run locally or through low-cost services where appropriate.

Layer C: Primary generative call

Use one main model call for context-sensitive dialogue or narration.

Layer D: Conditional secondary validation

Only invoke a more expensive validation model when:

- The conversation concerns safety-critical procedures
- The generated response references restricted facts
- The confidence of a lightweight checker is low
- The user asks an unusually open-ended question
- The scenario enters a high-risk state
- The output contains unsupported occupational claims

Most routine dialogue should not require multiple large-model calls.

Per-turn service-level targets

The team should define targets for:

- Median response latency
- 95th-percentile response latency
- Cost per generated interaction
- Maximum cost per completed simulation
- Percentage of turns requiring secondary review
- Safety-check failure rate
- Rate of blocked valid responses

Suggested initial product targets should be established through prototyping rather than assumed in advance.

Cost telemetry

Each session should record:

- Number of deterministic interactions
- Number of AI-generated interactions
- Number of secondary validation calls
- Tokens processed
- Cache-hit rate
- Moderation cost
- Generation cost
- Validation cost
- Total infrastructure cost

This allows the team to identify whether the safety architecture is commercially sustainable before scaling.

Graceful fallback

If validation services are unavailable or slow, the system should:

- Use an approved prewritten response
- Pause open-ended dialogue
- Continue deterministic interactions
- Avoid showing unvalidated AI output

Safety should not be bypassed to preserve responsiveness.

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4. Multi-NPC Scene and Dialogue Arbitration

Some scenes will contain several characters who have different knowledge, goals, relationships, and access to information.

Examples include:

- A manager and senior analyst in a meeting
- A physician, nurse, and patient in one room
- A teacher, student, and parent conference
- A designer presenting to several stakeholders

A single shared prompt containing every character’s private information would undermine NPC knowledge isolation.

Scene architecture

Multi-character scenes should use a scene coordinator.

```
Player action
↓
Scene coordinator identifies relevant participants
↓
Turn-arbitration rules select who may respond
↓
Separate knowledge context is assembled for each speaker
↓
NPC response is generated
↓
Public scene memory is updated
↓
Other NPCs receive only information spoken or observed publicly
```

Public and private memory

The system should distinguish:

Public scene memory

Information all present participants could reasonably observe or hear.

Private NPC memory

Information known only through:

- Earlier private conversations
- Role-specific systems
- Confidential documents
- Personal experience
- Restricted permissions

Shared subgroup memory

Information known to some, but not all, characters.

For example, two nurses may know something from a prior handoff that the patient does not know.

Turn arbitration

The scene coordinator should decide:

- Which NPC has reason to speak
- Whether multiple NPCs should respond
- Whether one NPC interrupts another
- Whether an NPC remains silent

- Which facts each response may reference
- What newly spoken facts become public

The coordinator should be rule-based where possible.

Separate generation calls

When two NPCs speak, each should normally receive an independently scoped context.

The second NPC may receive:

- The first NPC's publicly spoken line
- Publicly observable reactions
- Shared scene state

The second NPC should not receive the first NPC's private reasoning or restricted knowledge.

Scene narrator

A separate narrator component may describe:

- Body language
- Room changes
- General reactions
- Who is speaking
- Publicly visible consequences

The narrator should not expose private thoughts unless the product intentionally uses an omniscient presentation style.

Multi-NPC testing

Adversarial tests should verify that:

- NPCs do not reveal private conversations
- One NPC does not adopt another NPC's knowledge
- Confidential facts do not become public without being spoken or exposed
- Characters react only to information they could observe
- Meeting dialogue remains coherent across several turns

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5. Stochastic Replay and Regression Testing

Replay testing should not compare generated dialogue through exact text equality.

Language-model output is stochastic. Two valid runs may use different wording while preserving the same occupational meaning and structural outcome.

What replay tests should compare

Regression testing should focus on:

- World-state transitions
- Triggered events
- Task completion status
- Consequence timing
- NPC knowledge boundaries
- Professional-rule compliance
- Safety invariants
- Referenced fact identifiers
- Evidence records
- Prohibited outcome violations

Semantic output testing

Dialogue can be tested for semantic properties such as:

- Whether it communicates the required information
- Whether it contradicts an approved fact
- Whether it reveals restricted knowledge
- Whether it changes a procedure
- Whether its tone falls within the permitted range
- Whether it introduces an unsupported claim

Multi-sample testing

Critical prompts should be run multiple times.

For example:

```
Scenario checkpoint
→ Run 20 generations
→ Measure invariant violation rate
→ Measure leakage rate
→ Measure contradiction rate
→ Review semantic variation
```

A single successful generation does not establish reliability.

Regression thresholds

Each career pack should define acceptable thresholds for:

- State consistency

- Leakage
- Unsupported claims
- Moderation failures
- Schema failures
- Occupational inaccuracies

Safety-critical scenarios should use stricter thresholds than low-risk conversational scenes.

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6. Age-Tiered Privacy and Consent

The privacy model should vary according to the user's age, legal jurisdiction, institutional context, and account type.

The product should not assume that every user has identical rights or consent capacity.

Specific age thresholds and consent requirements must be established with qualified legal counsel for each region in which the platform operates.

Tier A: Young-child accounts

For users below the legally applicable consent threshold:

- Accounts should be created or authorized by a parent or guardian where required
- Data collection should be minimized
- Research participation should require separate appropriate consent
- Parent or guardian access and deletion mechanisms should follow applicable law
- Open-ended social features should be more restricted
- Sensitive reflection collection should be limited
- Personalized retention should default to the minimum needed

The product should clearly distinguish required guardian access from optional sharing.

Tier B: Teen accounts

Teen users should receive increasing control over:

- Reflections
- Saved simulations
- Counselor sharing
- Parent or guardian sharing
- Career lists
- Profile corrections

The exact balance between user privacy and guardian rights will depend on applicable law and institutional policy.

Tier C: Adult accounts

Adult users should directly control:

- Consent
- Data retention
- Sharing
- Export
- Deletion
- Research participation

School-managed accounts

Institutional accounts need separate policies governing:

- What the school can access
- What remains private to the student
- Whether counselors see raw reflections or summaries
- Whether simulation data becomes part of an educational record
- Data-retention periods
- Parent or guardian rights
- Use for research or product improvement

Data-access matrix

The product should define a formal access matrix by:

- Age tier
- Account type
- Data category
- User role
- Consent status
- Jurisdiction

This is safer than relying on broad statements such as “the user controls who sees their information.”

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7. Dynamic Career-Pathway Data Workstream

Career-pathway information should not be managed like static simulation content.

Information such as the following changes regularly:

- Salary ranges
- Licensing requirements
- Certification requirements
- Education pathways

- Job-market demand
- Credential costs
- Apprenticeship rules
- Geographic restrictions
- Continuing-education requirements
- Immigration or work-authorization requirements

It is also often jurisdiction-specific.

Separate pathway-data service

The platform should maintain a dedicated pathway-data layer containing:

```
Career
Region
Jurisdiction
Credential type
Education requirement
Typical training duration
Estimated cost
Licensing authority
Compensation source
Employment outlook source
Last verified date
Next review date
```

Update frequency

Different data types should have different update schedules.

For example:

- Licensing and legal requirements: checked whenever the responsible authority changes them and reviewed regularly
- Compensation data: updated at least annually where reliable sources exist
- Job-market indicators: updated according to source availability
- Education costs: reviewed by institution or region
- Certification requirements: checked against issuing organizations

Regional presentation

The platform should not show one pathway as universally applicable.

Instead:

> Requirements shown are for registered nurses in Ohio, United States. Requirements may differ in other jurisdictions.

Staleness policy

If information has not been verified within its required update period:

- Display a warning
- Avoid presenting it as current
- Direct users to the relevant official authority
- Prevent stale data from being used in strong recommendations

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8. Kill, Pivot, and Scope-Reduction Criteria

The project should define evidence that would challenge the core product hypothesis.

Positive user enjoyment alone should not justify continued expansion.

Core hypothesis

> A bounded immersive simulation improves career understanding and decision quality beyond simpler career-exploration materials enough to justify its additional complexity and cost.

Potential disconfirming results

The team should reconsider or pivot if:

- Users do not understand careers better than after reading high-quality career materials
- Practitioners consistently rate simulations as unrepresentative
- Debrief observations cannot be made reliably
- Simulation behavior is dominated by interface skill or gaming familiarity
- Occupational accuracy requires unsustainable authoring cost
- AI inconsistency remains too high after bounded-generation controls
- Infrastructure costs exceed viable session economics
- Accessibility modes produce materially different conclusions
- Users experience increased confidence without increased accuracy
- Career recommendations create harmful or biased steering

Possible responses

Redesign

Use when the core concept appears valuable but implementation is causing failure.

Examples:

- Simplify navigation
- Reduce AI generation
- Increase authored structure
- Narrow debrief claims
- Improve occupational content

Scope reduction

Use when only part of the platform proves useful.

Examples:

- Use simulations only as discussion starters
- Remove skill-performance claims
- Focus on job realism rather than career fit
- Serve counselors rather than direct-to-student users
- Limit the platform to certain career categories

Pivot

Use when immersive interaction is not producing additional decision value.

Possible pivots include:

- Interactive day-in-the-life storytelling
- Counselor-led simulation workshops
- Career-comparison modules
- Job-shadow preparation tools
- Practitioner-authored scenario libraries

Halt

Expansion should stop if the platform repeatedly produces misleading guidance, cannot meet safety or privacy requirements, or shows no meaningful advantage over much simpler alternatives.

Stage-gate decision

At the Phase 1.5 checkpoint, the team should formally decide:

```
Proceed
Proceed with narrower claims
Redesign and retest
Pivot
Halt
```

The criteria for each decision should be documented before results are reviewed to reduce confirmation bias.

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9. Scenario-Rhythm Bias Review

Bias review should evaluate not only what occurs within a scenario, but also which scenario rhythms are assigned to each career.

A career may be stereotyped before any dialogue is written.

Examples:

- Nursing represented only through emotional caregiving
- Teaching represented only through social conflict
- Human resources represented only through relationship management
- Cybersecurity represented only through emergencies
- Engineering represented only through solitary analysis
- Skilled trades represented only through physical procedure

Rhythm-diversity requirement

Each career pack should document the range of authentic work rhythms within the occupation.

Where supported by occupational evidence, packs should include more than one rhythm.

Example:

```
Cybersecurity analyst:
- Routine monitoring
- Quiet investigation
- Team coordination
- Operational incident response

Teacher:
- Focused lesson planning
- Classroom facilitation
- Administrative work
- Student support
- Parent communication
```

Bias-review checklist

Reviewers should ask:

- Does the selected rhythm reflect ordinary work or only culturally familiar imagery?
- Are calm and repetitive parts represented?
- Are technical and relational aspects both represented where appropriate?
- Is emotional labor overemphasized in historically feminized professions?
- Is crisis response overemphasized in historically male-coded technical professions?
- Does the career pack show variation across settings and specializations?
- Would practitioners from different backgrounds recognize the portrayal?

Reason

Scenario structure itself communicates what the platform believes the occupation fundamentally is. Rhythm selection must therefore be treated as a content and fairness decision, not merely a narrative-design choice.